



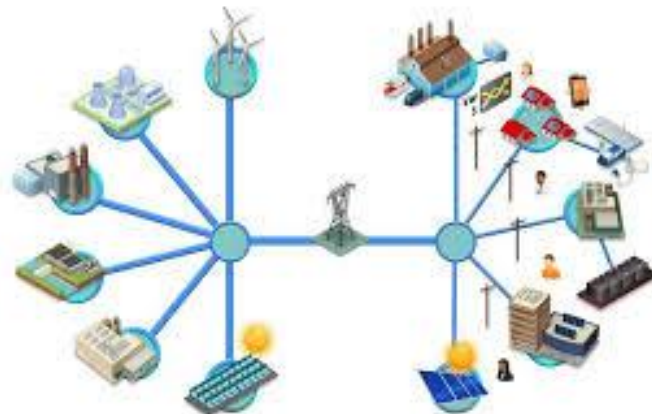
NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Grid access

Nodal Prices and Optimal Power Flow

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Learning outcome

- Lagrange multipliers – short repetition
- OPF equations based on Lagrange multiplier approach
- Example using 3-node DC network with resistances
- Nodal prices, transmission losses, loss factors, security of supply

How to derive Lagrangian function?

A linear program is in **standard form** if it is written as

Minimize $f(x)$
 x

subject to:

$$h(x) = 0 \quad : \quad \lambda$$

$$g(x) \leq 0 \quad : \quad \mu$$



$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^\top h(x) + \mu^\top g(x)$$

How to derive optimality conditions?

Original (primal) problem

$$\begin{aligned} &\underset{x}{\text{Minimize}} && f(x) \\ &\text{subject to:} && \\ &h(x) = 0 && : \lambda \\ &g(x) \leq 0 && : \mu \end{aligned}$$



Lagrangian function

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \lambda^\top h(x) + \mu^\top g(x)$$

Optimality Karush–Kuhn–Tucker (KKT) conditions

$$\begin{aligned} &\frac{\partial \mathcal{L}(x, \lambda, \mu)}{\partial x} = 0 \\ &h(x) = 0 \\ &g(x) \leq 0 \\ &\mu \geq 0 \\ &\mu g(x) = 0 \\ &\lambda \in \text{free} \end{aligned}$$



$$\begin{aligned} &\frac{\partial \mathcal{L}(x, \lambda, \mu)}{\partial x} = 0 \\ &h(x) = 0 \\ &0 \leq -g(x) \perp \mu \geq 0 \\ &\lambda \in \text{free} \end{aligned}$$

Complementarity
Slackness Condition

Lagrange multipliers

$$\min J = \min f(x_1, x_2, \dots, x_n)$$

subject to :

$$g(x_1, x_2, \dots, x_n) = 0$$



$$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda g(x_1, x_2, \dots, x_n)$$

λ : Lagrange multipliers

Optimality Karush–Kuhn–Tucker (KKT) conditions

Derivate wrt X and λ :

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1} = 0$$

...

$$\frac{\partial \mathcal{L}}{\partial x_n} = \frac{\partial f}{\partial x_n} + \lambda \frac{\partial g}{\partial x_n} = 0$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = g(x_1, x_2, \dots, x_n) = 0$$

$$\begin{aligned} \frac{\partial \mathcal{L}(x, \lambda)}{\partial x} &= 0 \\ \frac{\partial \mathcal{L}(x, \lambda)}{\partial \lambda} &= g(x) = 0 \end{aligned}$$

➤ In optimum:

- constraints satisfied
- $\min L = \min f$

Power systems – meaning of λ

- Market Clearing with inelastic demand (Centralized social welfare optimization)

$$\begin{aligned} \max_{\{P^G\}, \{P^D\}} \quad & \sum_d U_d^D P_d^D - \sum_g c_g^G P_g^G \\ \text{subject to} \quad & \sum_d P_d^D - \sum_g P_g^G = 0 \quad (\lambda) \\ & 0 \leq P_d^D \leq \bar{P}_d^D, d=1, \dots, N_D \\ & 0 \leq P_g^G \leq \bar{P}_g^G, g=1, \dots, N_G \end{aligned}$$

$$\begin{aligned} \min_{\{P^G\}} \quad & \sum_g c_g^G P_g^G \\ \text{subject to} \quad & D - \sum_g P_g^G = 0 \quad (\lambda) \\ & 0 \leq P_g^G \leq \bar{P}_g^G, g=1, \dots, N_G \end{aligned}$$

- Minimize f: generation cost functions

$$f = c_1 P_1 + c_2 P_2 + \dots + c_n P_n$$

- Simplest case:
 - one constraint:

$$e = D - \sum_g P_g = 0 \quad (\lambda)$$

- In optimum:

$$f + \lambda \left(D - \sum_g P_g \right) \left\{ \begin{array}{l} \frac{\partial L}{\partial P_1} = \frac{\partial f}{\partial P_1} + \lambda \frac{\partial e}{\partial P_1} = 0 \\ \dots \\ \frac{\partial L}{\partial P_n} = \frac{\partial f}{\partial P_n} + \lambda \frac{\partial e}{\partial P_n} = 0 \end{array} \right\} \Rightarrow \frac{\frac{\partial f}{\partial P_1}}{-\frac{\partial e}{\partial P_1}} = \frac{\frac{\partial f}{\partial P_2}}{-\frac{\partial e}{\partial P_2}} = \dots = \frac{\frac{\partial f}{\partial P_n}}{-\frac{\partial e}{\partial P_n}} = \lambda$$

- In this case all $\frac{\partial e}{\partial P_g} = -1$ and all $\frac{\partial f}{\partial P_g} = \lambda$

Power systems – meaning of λ

- General case
$$\frac{\frac{\partial f}{\partial P_1}}{\frac{\partial e}{\partial P_1}} = \frac{\frac{\partial f}{\partial P_2}}{\frac{\partial e}{\partial P_2}} = \dots = \frac{\frac{\partial f}{\partial P_n}}{\frac{\partial e}{\partial P_n}} = \lambda$$
- Ratio between derivatives of “ f ” and “ e ” equal for all $P_g \rightarrow (\lambda)$
- **Numerator:**
 - the rate for an increase of the objective function by a marginal increase of $P_g =$ additional cost per additional unit of P_g .
- **Denominator:**
 - impact of a change in the constraint
- λ represents the marginal cost for an adjustment of the constraint
 - λ ties together the derivatives of the costs and constraints
 - λ is also known as “shadow price”

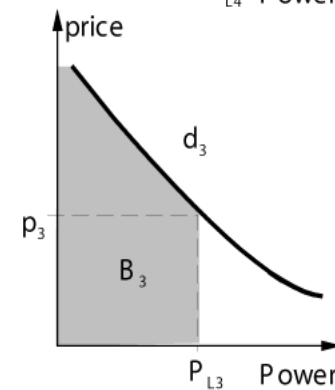
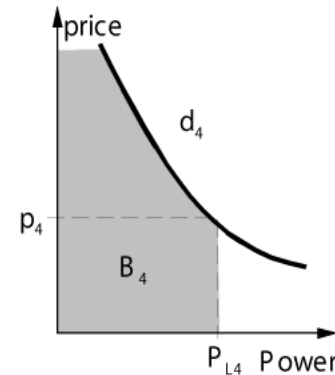
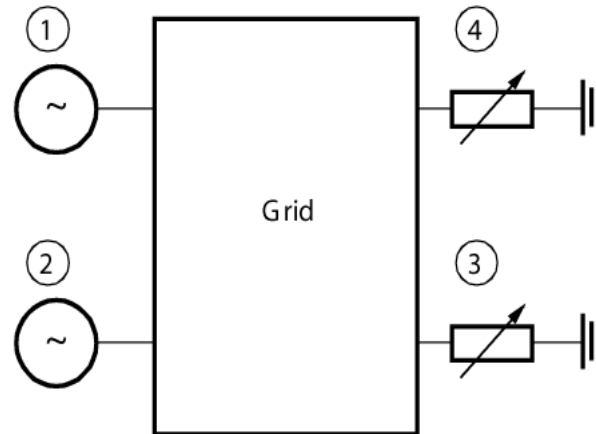
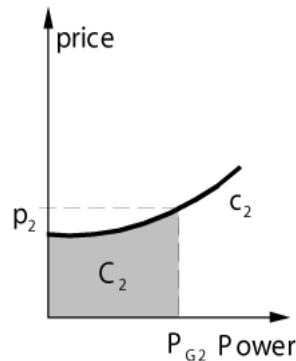
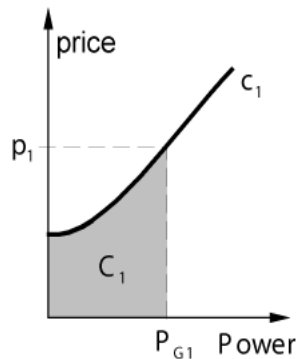
Nodal prices based on OPF

$$c_i = \frac{\partial C_i}{\partial P_{Gi}} = c_i(P_{Gi})$$

C_i is the cost of generation in point i
 c_i is the marginal cost

$$d_j = \frac{\partial B_j}{\partial P_{Lj}} = d_j(P_{Lj})$$

B_j is the total utility for consumer j
 d_j is the marginal willingness to pay



Unconstrained case

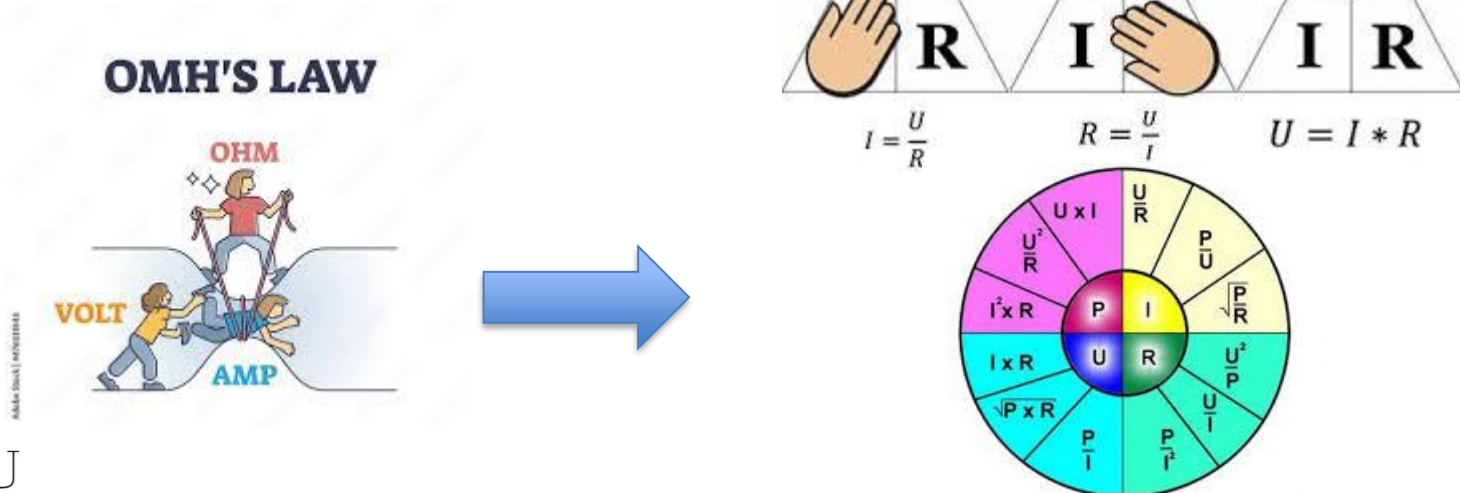
$$\lambda = \begin{bmatrix} c \\ d \end{bmatrix}$$

Several Grid Models

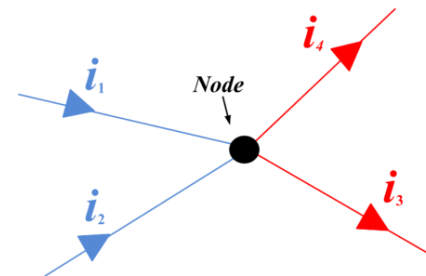
OPF models	
"Transport model" (Non-electrical model)	• Grid modeled with ATC's • Normally not called OPF • Nord Pool (before November 2024)
DC model	• DC grid • Not used for real grids, but handy to demonstrate principles • Straight forward to include losses
DC-equivalent model "DC Power Flow"	• Approximate model of AC grid • Reasonable approximation of active flow • Used many places • Linear model
AC model	• Correct description but computationally more intensive

Why DC example?

- **DC grid**, not DC load flow in AC grid!
- Slightly easier to understand load flow equations
- Principal approach is very similar



Problem Formulation



$$\min f(\mathbf{P})$$

: Objective Function

$\mathbf{P}, \mathbf{U}$ → state vector, i.e. the voltage at every node
 → input/output vector

St.

$$g(\mathbf{U}, \mathbf{P}) = 0 \quad (\boldsymbol{\lambda}) \quad \text{: Power flow Equations (one constraint for each node! ☺)}$$

$$h(\mathbf{U}) \leq 0 \quad (\boldsymbol{\eta}) \quad \text{: Transmission Capacity}$$

$$\mathbf{P} \leq \mathbf{P}_{\max} \rightarrow \mathbf{P} - \mathbf{P}_{\max} \leq 0 \quad (\boldsymbol{\gamma}) \quad \text{: Generation Capacity}$$

Lagrange function:

$$\mathcal{L}(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}, \boldsymbol{\eta}, \boldsymbol{\gamma}) = f(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{U}, \mathbf{P}) + \boldsymbol{\eta}^t \mathbf{h}(\mathbf{U}) + \boldsymbol{\gamma}^t (\mathbf{P} - \mathbf{P}_{\max})$$

First order Karush–Kuhn–Tucker (KKT) optimality conditions

$$L(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}, \boldsymbol{\eta}, \boldsymbol{\gamma}) = f(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{U}, \mathbf{P}) + \boldsymbol{\eta}^t \mathbf{h}(\mathbf{U}) + \boldsymbol{\gamma}^t (\mathbf{P} - \mathbf{P}_{\max})$$

$$\frac{\partial L}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \boldsymbol{\lambda} + \left(\frac{\partial \mathbf{h}}{\partial \mathbf{U}} \right)^t \boldsymbol{\eta} = 0$$

$$\frac{\partial L}{\partial \mathbf{P}} = 0 \Rightarrow \frac{\partial f}{\partial \mathbf{P}} + \left(\frac{\partial \mathbf{g}}{\partial \mathbf{P}} \right)^t \boldsymbol{\lambda} + \left(\frac{\partial (\mathbf{P} - \mathbf{P}_{\max})}{\partial \mathbf{P}} \right)^t \boldsymbol{\gamma} = 0$$

$$\frac{\partial L}{\partial \boldsymbol{\lambda}} = 0 \Rightarrow \mathbf{g}(\mathbf{U}, \mathbf{P}) = 0$$

$$\frac{\partial L}{\partial \boldsymbol{\eta}} = 0 \Rightarrow \mathbf{h}(\mathbf{U}) = 0$$

$$\frac{\partial L}{\partial \boldsymbol{\gamma}} = 0 \Rightarrow (\mathbf{P} - \mathbf{P}_{\max}) = 0$$

Complementary Slackness Conditions:

$$\boldsymbol{\eta}^{*t} \mathbf{h}(\mathbf{U}^*) = 0$$

$$\mathbf{h}(\mathbf{U}^*) \leq 0$$

$$\boldsymbol{\eta}^* \geq 0$$

$$\boldsymbol{\gamma}^{*t} (\mathbf{P}^* - \mathbf{P}_{\max}) = 0$$

$$\mathbf{P}^* - \mathbf{P}_{\max} \leq 0$$

$$\boldsymbol{\gamma}^* \geq 0$$

KKT conditions, no limitation on generation and transmission capacity

$$L(\mathbf{U}, \mathbf{P}, \lambda, \boldsymbol{\eta}, \boldsymbol{\gamma}) = f(\mathbf{P}) + \lambda^t \mathbf{g}(\mathbf{U}, \mathbf{P}) + \cancel{\boldsymbol{\eta}^t \mathbf{h}(\mathbf{U})} + \cancel{\boldsymbol{\gamma}^t (\mathbf{P} - \mathbf{P}_{\max})}$$

$$\Rightarrow L(\mathbf{U}, \mathbf{P}, \lambda) = f(\mathbf{P}) + \lambda^t \mathbf{g}(\mathbf{U}, \mathbf{P})$$

Derivate $L(\mathbf{U}, \mathbf{P}, \lambda)$ wrt λ , $\mathbf{U}$ and $\mathbf{P}$:

$$\frac{\partial L}{\partial \lambda} = 0 \Rightarrow \mathbf{g}(\mathbf{U}, \mathbf{P}) = 0$$

$$\frac{\partial L}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \lambda = 0$$

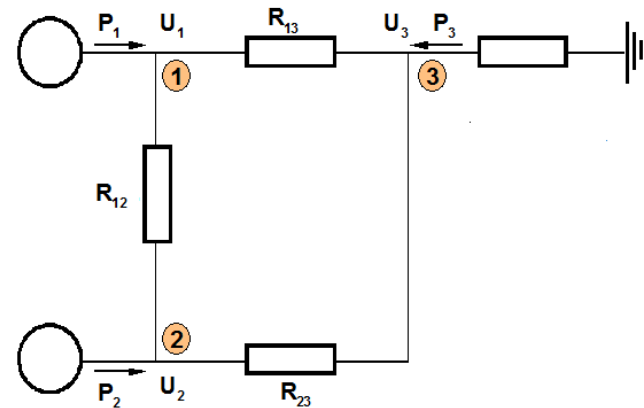
$$\frac{\partial L}{\partial \mathbf{P}} = \frac{\partial \mathbf{f}}{\partial \mathbf{P}} + \left(\frac{\partial \mathbf{g}}{\partial \mathbf{P}} \right)^t \lambda = 0$$

3-node DC grid, uncongested case without generation capacity limitation

Lagrange function:

$$L(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}) = \mathbf{f}(\mathbf{P}) + \boldsymbol{\lambda}' \mathbf{g}(\mathbf{U}, \mathbf{P}) = f_1(P_1) + f_2(P_2) + f_3(P_3) + \lambda_1 g_1(\mathbf{U}, \mathbf{P}) + \lambda_2 g_2(\mathbf{U}, \mathbf{P}) + \lambda_3 g_3(\mathbf{U}, \mathbf{P})$$

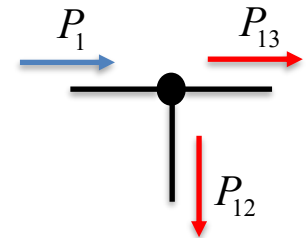
$$\mathbf{P} = \begin{bmatrix} P_1 \\ P_2 \\ -P_3 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix}, \quad \boldsymbol{\lambda} = \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix}$$



Load flow equation for node 1:

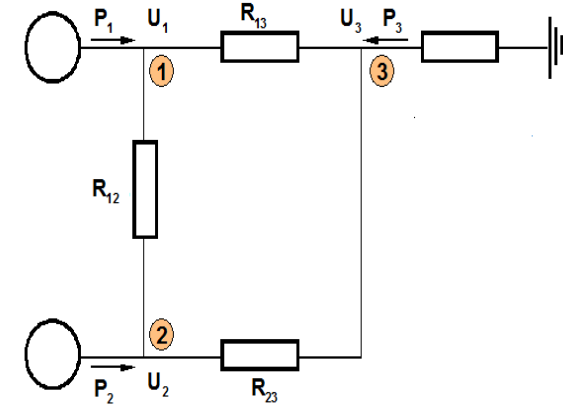
$$g_1(\mathbf{U}, \mathbf{P}) : P_1 = P_{12} + P_{13} = I_{12}U_1 + I_{13}U_1 = \left(\frac{U_1 - U_2}{R_{12}} \right) U_1 + \left(\frac{U_1 - U_3}{R_{13}} \right) U_1 \Rightarrow$$

$$g_1(\mathbf{U}, \mathbf{P}) = -P_1 + \left[\frac{(U_1^2 - U_1U_2)}{R_{12}} + \frac{(U_1^2 - U_1U_3)}{R_{13}} \right] = 0$$



Load flow equations

$$\mathbf{g}(\mathbf{U}, \mathbf{P}) = 0 \rightarrow \begin{cases} g_1(\mathbf{U}, \mathbf{P}) = -P_1 + \left[\frac{(U_1^2 - U_1 U_2)}{R_{12}} + \frac{(U_1^2 - U_1 U_3)}{R_{13}} \right] = 0 \\ g_2(\mathbf{U}, \mathbf{P}) = -P_2 + \left[\frac{(U_2^2 - U_2 U_1)}{R_{12}} + \frac{(U_2^2 - U_2 U_3)}{R_{23}} \right] = 0 \\ g_3(\mathbf{U}, \mathbf{P}) = -P_3 + \left[\frac{(U_3^2 - U_3 U_1)}{R_{13}} + \frac{(U_3^2 - U_3 U_2)}{R_{23}} \right] = 0 \end{cases}$$



$$\mathbf{L}(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}) = \mathbf{f}(\mathbf{P}) + \boldsymbol{\lambda}^t \mathbf{g}(\mathbf{U}, \mathbf{P})$$

$$L(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}) = 0 \Rightarrow \begin{bmatrix} c_1 & c_2 & c_3 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} + \begin{bmatrix} \lambda_1 & \lambda_2 & \lambda_3 \end{bmatrix} \begin{bmatrix} -P_1 + \left(\frac{(U_1^2 - U_1 U_2)}{R_{12}} + \frac{(U_1^2 - U_1 U_3)}{R_{13}} \right) \\ -P_2 + \left(\frac{(U_2^2 - U_2 U_1)}{R_{12}} + \frac{(U_2^2 - U_2 U_3)}{R_{23}} \right) \\ -P_3 + \left(\frac{(U_3^2 - U_3 U_1)}{R_{13}} + \frac{(U_3^2 - U_3 U_2)}{R_{23}} \right) \end{bmatrix} = 0$$

Define Jacobian matrix (derivative of vector wrt vector)

Derivate of **g** wrt **U**: $\frac{\partial \mathbf{L}}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \lambda = 0$

$$\mathbf{g} = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix} \quad \text{and} \quad \mathbf{U} = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix}$$

$$\frac{\partial \mathbf{g}}{\partial \mathbf{U}} = \begin{bmatrix} \frac{\partial g_1}{\partial U_1} & \frac{\partial g_1}{\partial U_2} & \frac{\partial g_1}{\partial U_3} \\ \frac{\partial g_2}{\partial U_1} & \frac{\partial g_2}{\partial U_2} & \frac{\partial g_2}{\partial U_3} \\ \frac{\partial g_3}{\partial U_1} & \frac{\partial g_3}{\partial U_2} & \frac{\partial g_3}{\partial U_3} \end{bmatrix} \rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t = \begin{bmatrix} \frac{\partial g_1}{\partial U_1} & \frac{\partial g_2}{\partial U_1} & \frac{\partial g_3}{\partial U_1} \\ \frac{\partial g_1}{\partial U_2} & \frac{\partial g_2}{\partial U_2} & \frac{\partial g_3}{\partial U_2} \\ \frac{\partial g_1}{\partial U_3} & \frac{\partial g_2}{\partial U_3} & \frac{\partial g_3}{\partial U_3} \end{bmatrix}$$

Derivate of Lagrange function wrt U :

Load Flow Equations

$$g_1(\mathbf{U}, \mathbf{P}) = -P_1 + \left[\frac{(U_1^2 - U_1 U_2)}{R_{12}} + \frac{(U_1^2 - U_1 U_3)}{R_{13}} \right] = 0$$

$$g_2(\mathbf{U}, \mathbf{P}) = -P_2 + \left[\frac{(U_2^2 - U_2 U_1)}{R_{12}} + \frac{(U_2^2 - U_2 U_3)}{R_{23}} \right] = 0$$

$$g_3(\mathbf{U}, \mathbf{P}) = -P_3 + \left[\frac{(U_3^2 - U_3 U_1)}{R_{13}} + \frac{(U_3^2 - U_3 U_2)}{R_{23}} \right] = 0$$

$$\frac{\partial \mathbf{L}}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \boldsymbol{\lambda} = 0$$

$$\frac{\partial \mathbf{L}}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \boldsymbol{\lambda} = 0 \Rightarrow \underbrace{\begin{bmatrix} \frac{\partial g_1}{\partial U_1} & \frac{\partial g_2}{\partial U_1} & \frac{\partial g_3}{\partial U_1} \\ \frac{\partial g_1}{\partial U_2} & \frac{\partial g_2}{\partial U_2} & \frac{\partial g_3}{\partial U_2} \\ \frac{\partial g_1}{\partial U_3} & \frac{\partial g_2}{\partial U_3} & \frac{\partial g_3}{\partial U_3} \end{bmatrix}}_{\text{Jacobian matrix}} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} \lambda_1 \frac{\partial g_1}{\partial U_1} + \lambda_2 \frac{\partial g_2}{\partial U_1} + \lambda_3 \frac{\partial g_3}{\partial U_1} \\ \lambda_1 \frac{\partial g_1}{\partial U_2} + \lambda_2 \frac{\partial g_2}{\partial U_2} + \lambda_3 \frac{\partial g_3}{\partial U_2} \\ \lambda_1 \frac{\partial g_1}{\partial U_3} + \lambda_2 \frac{\partial g_2}{\partial U_3} + \lambda_3 \frac{\partial g_3}{\partial U_3} \end{bmatrix} = 0$$

KKT conditions, no limitation on generation capacity

$$\mathbf{L}(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}) = \mathbf{f}(\mathbf{P}) + \boldsymbol{\lambda}' \mathbf{g}(\mathbf{U}, \mathbf{P})$$

Derivate of Lagrange function wrt $\mathbf{P}$:

$$\frac{\partial \mathbf{L}}{\partial \mathbf{P}} = \frac{\partial \mathbf{f}}{\partial \mathbf{P}} + \left(\frac{\partial \mathbf{g}}{\partial \mathbf{P}} \right)' \boldsymbol{\lambda} = 0 \Rightarrow$$

$$\begin{bmatrix} \frac{\partial f_1}{\partial P_1} \\ \frac{\partial f_2}{\partial P_2} \\ \frac{\partial f_3}{\partial P_3} \end{bmatrix} + \begin{bmatrix} \frac{\partial g_1}{\partial P_1} & \frac{\partial g_2}{\partial P_1} & \frac{\partial g_3}{\partial P_1} \\ \frac{\partial g_1}{\partial P_2} & \frac{\partial g_2}{\partial P_2} & \frac{\partial g_3}{\partial P_2} \\ \frac{\partial g_1}{\partial P_3} & \frac{\partial g_2}{\partial P_3} & \frac{\partial g_3}{\partial P_3} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = 0 \quad \text{and} \quad \begin{bmatrix} c_1 \\ c_2 \\ d_3 \end{bmatrix} + \begin{bmatrix} -\lambda_1 \\ -\lambda_2 \\ -\lambda_3 \end{bmatrix} = 0 \Rightarrow \boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} \\ \mathbf{d} \end{bmatrix}$$

Because $\frac{\partial f}{\partial P}$ are marginal generation costs and $\frac{\partial g}{\partial P} = -1$

KKT conditions, with limitation on generation capacity

$$L(\mathbf{U}, \mathbf{P}, \boldsymbol{\lambda}, \boldsymbol{\gamma}) = \mathbf{f}(\mathbf{P}) + \boldsymbol{\lambda}' \mathbf{g}(\mathbf{U}, \mathbf{P}) + \boldsymbol{\gamma}' (\mathbf{P} - \mathbf{P}_{\max}) = 0$$

Derivate of Lagrange function wrt $\mathbf{P}$:

$$\frac{\partial L}{\partial \mathbf{P}} = \frac{\partial \mathbf{f}}{\partial \mathbf{P}} + \left(\frac{\partial \mathbf{g}}{\partial \mathbf{P}} \right)^t \boldsymbol{\lambda} + \left(\frac{\partial (\mathbf{P} - \mathbf{P}_{\max})}{\partial \mathbf{P}} \right)^t \boldsymbol{\gamma} = 0 \Rightarrow$$

$$\begin{bmatrix} \frac{\partial f_1}{\partial P_1} \\ \frac{\partial f_2}{\partial P_2} \\ \frac{\partial f_3}{\partial P_3} \end{bmatrix} + \begin{bmatrix} \frac{\partial g_1}{\partial P_1} & \frac{\partial g_2}{\partial P_1} & \frac{\partial g_3}{\partial P_1} \\ \frac{\partial g_1}{\partial P_2} & \frac{\partial g_2}{\partial P_2} & \frac{\partial g_3}{\partial P_2} \\ \frac{\partial g_1}{\partial P_3} & \frac{\partial g_2}{\partial P_3} & \frac{\partial g_3}{\partial P_3} \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} + \begin{bmatrix} \frac{\partial (P_1 - P_{1\max})}{\partial P_1} & 0 & 0 \\ 0 & \frac{\partial (P_2 - P_{2\max})}{\partial P_2} & 0 \\ 0 & 0 & \frac{\partial (P_3 - P_{3\max})}{\partial P_3} \end{bmatrix} \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} = 0$$

and

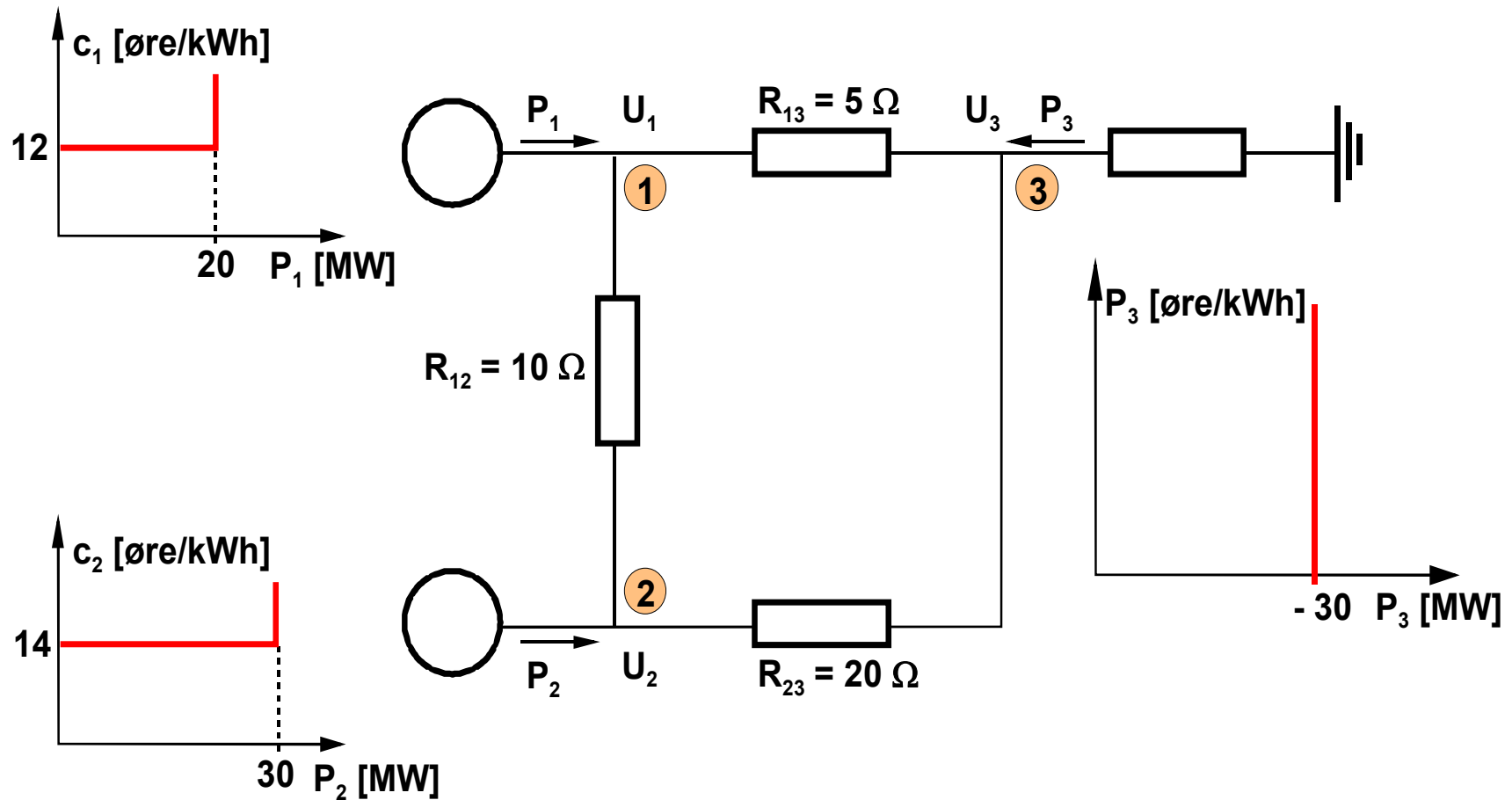
$$\begin{bmatrix} c_1 \\ c_2 \\ d_3 \end{bmatrix} + \begin{bmatrix} -\lambda_1 \\ -\lambda_2 \\ -\lambda_3 \end{bmatrix} + \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{bmatrix} = 0 \Rightarrow \boldsymbol{\lambda} = \begin{bmatrix} \mathbf{c} + \boldsymbol{\gamma} \\ \mathbf{d} \end{bmatrix}$$

Nodal prices, constrained case

$$\lambda = \begin{bmatrix} \mathbf{c} + \gamma \\ \mathbf{d} \end{bmatrix}$$

- λ_i marginal cost of supplying one marginal unit at bus i
- generation node: price = $\lambda_i = c_i + \gamma_i$
 - with infinite generation capacity at node i the price is always equal to the marginal cost of generation
 - γ is shadow cost of constraint on generation capacity
- load node: price = $\lambda_j = d_j$
 - because there is no limit on the load in this formulation
- with a fixed demand, there will also be a shadow cost in the load node
 - equality constraint, $D=D_{\text{fixed}}$, in this case $\lambda_i = \gamma_i$

Example (DC grid)



Three node example, uncongested case

Bus admittance matrix

$$\mathbf{Y} = \begin{bmatrix} 0.3 & -0.1 & -0.2 \\ -0.1 & 0.15 & -0.05 \\ -0.2 & -0.05 & 0.25 \end{bmatrix}$$

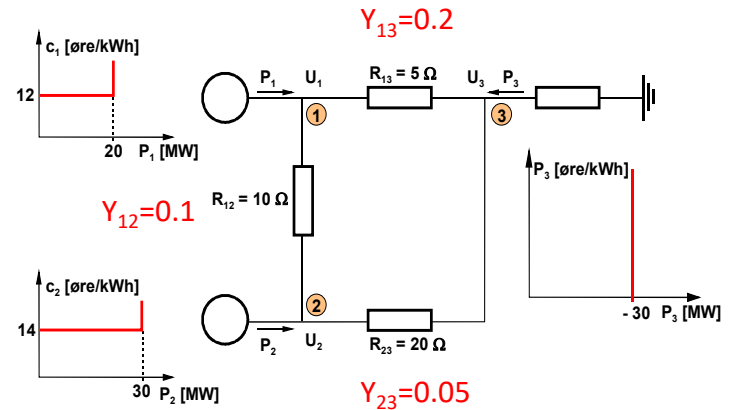
Current vector:

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \mathbf{YU} = \begin{bmatrix} 0.3U_1 - 0.1U_2 - 0.2U_3 \\ 0.1U_1 + 0.15U_2 - 0.05U_3 \\ 0.2U_1 - 0.05U_2 + 0.25U_3 \end{bmatrix}$$

Load flow equations:

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} I_1 U_1 \\ I_2 U_2 \\ I_3 U_3 \end{bmatrix} = \begin{bmatrix} 0.3U_1^2 - 0.1U_1U_2 - 0.2U_1U_3 \\ -0.1U_1U_2 + 0.15U_2^2 - 0.05U_2U_3 \\ -0.2U_1U_3 - 0.05U_2U_3 + 0.25U_3^2 \end{bmatrix}$$

$$\mathbf{g}(\mathbf{U}, \mathbf{P}) = 0 \Rightarrow \begin{bmatrix} -P_1 + 0.3U_1^2 - 0.1U_1U_2 - 0.2U_1U_3 \\ -P_2 - 0.1U_1U_2 + 0.15U_2^2 - 0.05U_2U_3 \\ -P_3 - 0.2U_1U_3 - 0.05U_2U_3 + 0.25U_3^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



Three node example, uncongested case

Choose voltage in reference bus 3: 60 kV.

Given load $P_3 = 30$ MW, $G_1 = 20$ MW (lowest cost, no grid constraints):

$$0.3U_1^2 - 0.1U_1U_2 - 0.2 \times 60 \times U_1 - 20 = 0$$

$$-0.1U_1U_2 + 0.15U_2^2 - 0.05 \times 60 \times U_2 - P_2 = 0$$

$$-0.2 \times 60 \times U_1 - 0.05 \times 60 \times U_2 + 0.25 \times 60^2 - (-30) = 0$$

$$U_1 = 61.89 \text{ kV}$$

$$U_2 = 62.44 \text{ kV}$$

from last equation: $U_2 = 310 - 4U_1$

Substitute in first equation and reorder:

$$\begin{bmatrix} U_1 \\ U_2 \\ U_3 \end{bmatrix} = \begin{bmatrix} 61.89 \\ 62.44 \\ 60.00 \end{bmatrix} [kV]$$

Using equation for P_2 : $P_2 = 11.05$ MW

$$P_{loss} = \sum_i P_{Gi} - Load = 31.05 - 30 = 1.05 \text{ MW}$$

Three node example, uncongested case

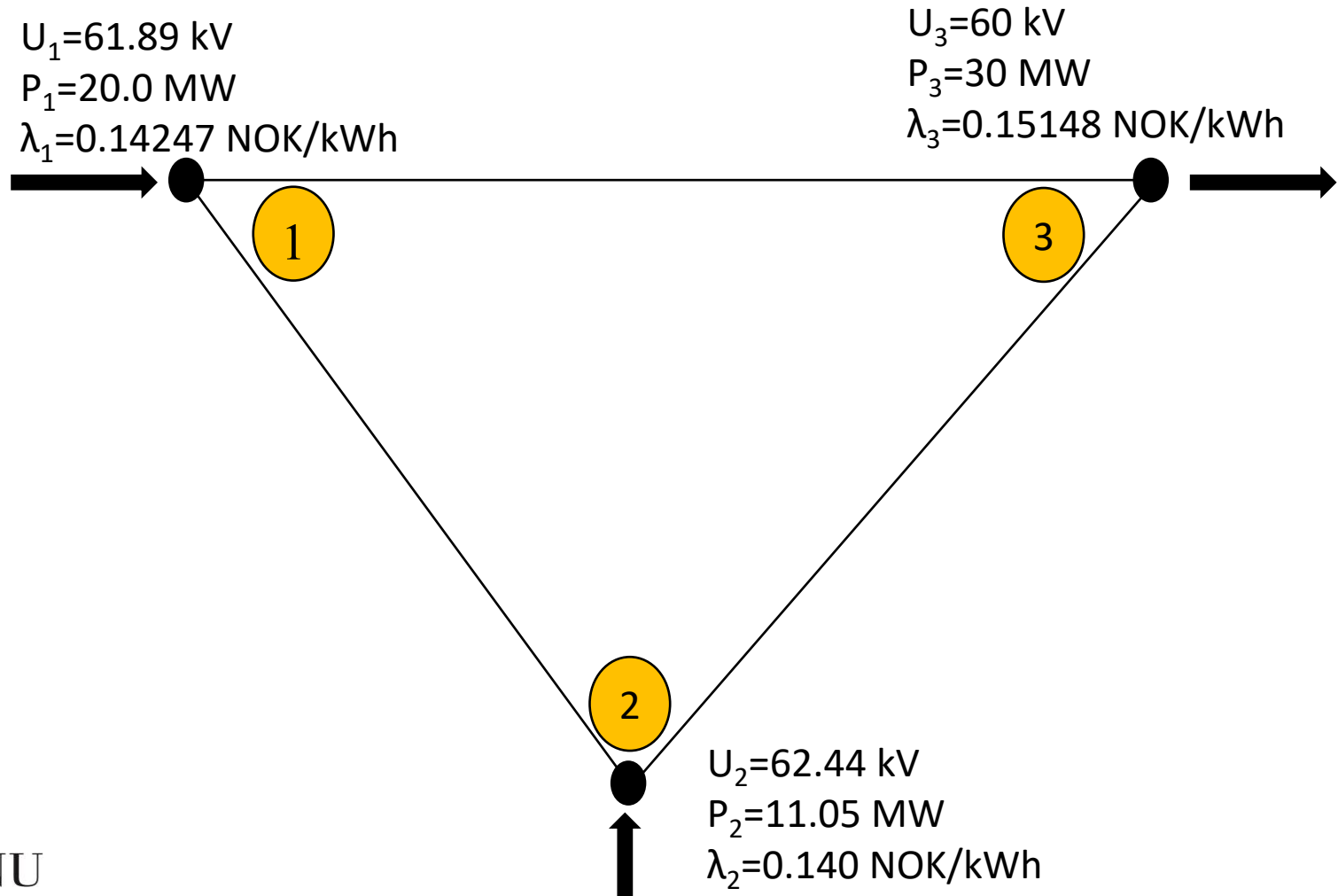
Nodal prices

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}}\right)^t \boldsymbol{\lambda} = 0 \quad \mathbf{g}(\mathbf{U}, \mathbf{P}) = 0 \quad \Rightarrow \quad \begin{bmatrix} -P + 0.3U_1^2 - 0.1U_1U_2 - 0.2U_1U_3 \\ -P - 0.1U_1U_2 + 0.15U_2^2 - 0.05U_2U_3 \\ -P_3 - 0.2U_1U_3 - 0.05U_2U_3 + 0.25U_3^2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

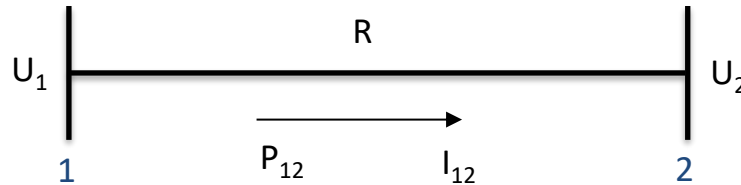
$$\frac{\partial \mathbf{g}}{\partial \mathbf{U}} = \begin{bmatrix} \frac{\partial g_1}{\partial U_1} & \frac{\partial g_1}{\partial U_2} \\ \frac{\partial g_2}{\partial U_1} & \frac{\partial g_2}{\partial U_2} \\ \frac{\partial g_3}{\partial U_1} & \frac{\partial g_3}{\partial U_2} \end{bmatrix} = \begin{bmatrix} 0.6U_1 - 0.1U_2 - 0.2U_3 & -0.1U_1 \\ -0.1U_2 & 0.3U_2 - 0.1U_1 - 0.05U_3 \\ -0.2U_3 & -0.05U_3 \end{bmatrix} = \begin{bmatrix} 18.89 & -6.189 \\ -6.244 & 9.543 \\ -12 & -3 \end{bmatrix}$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}}\right)^t \boldsymbol{\lambda} = \begin{bmatrix} 18.89\lambda_1 - 6.244\lambda_2 - 12.00\lambda_3 \\ -6.189\lambda_1 + 9.543\lambda_2 - 3.00\lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} c_1 + \gamma_1 \\ c_2 + \gamma_2 \\ \gamma_3 \end{bmatrix} = \begin{bmatrix} 14,247 \\ 14,0 \\ 15,148 \end{bmatrix} \text{ [øre/kWh]}$$

3-node DC grid, uncongested case



Transmission active losses



$$I_{12} = (U_1 - U_2) \frac{1}{R} \quad P_{12} = U_1 I_{12}$$

$$Loss_{12} = R I_{12}^2 = R \left(\frac{P_{12}}{U_1} \right)^2$$

$$\frac{\partial Loss_{12}}{\partial P_{12}} = 2 \frac{R \cdot P_{12}}{U_1^2} = 2 \left(1 - \frac{U_2}{U_1} \right)$$

If we transmit a small quantity ΔP_1 from node 1, and the corresponding quantity arriving at node 2 is ΔP_2 , then ρ_{12} is given by:

$$\rho_{12} = \frac{\partial Loss_{12}}{\partial P_{12}} = \frac{\Delta P_1 - \Delta P_2}{\Delta P_1} = 2 \left(1 - \frac{U_2}{U_1} \right)$$

$$\frac{U_2}{U_1} = 1 - \frac{\rho_{12}}{2}$$

Transmission active losses

Power flow equations in the two-node system:

$$\begin{bmatrix} g_1(\mathbf{U}, \mathbf{P}) \\ g_2(\mathbf{U}, \mathbf{P}) \end{bmatrix} = \begin{bmatrix} -P_1 + \left(\frac{U_1^2 - U_1 U_2}{R} \right) \\ -P_2 + \left(\frac{U_2^2 - U_2 U_1}{R} \right) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\frac{\partial \mathbf{L}}{\partial \mathbf{U}} = 0 \Rightarrow \left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \boldsymbol{\lambda} = 0 \Rightarrow \begin{bmatrix} \lambda_1 \left(\frac{2U_1 - U_2}{R} \right) + \lambda_2 \left(-\frac{U_2}{R} \right) \\ \lambda_1 \left(-\frac{U_1}{R} \right) + \lambda_2 \left(\frac{2U_2 - U_1}{R} \right) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

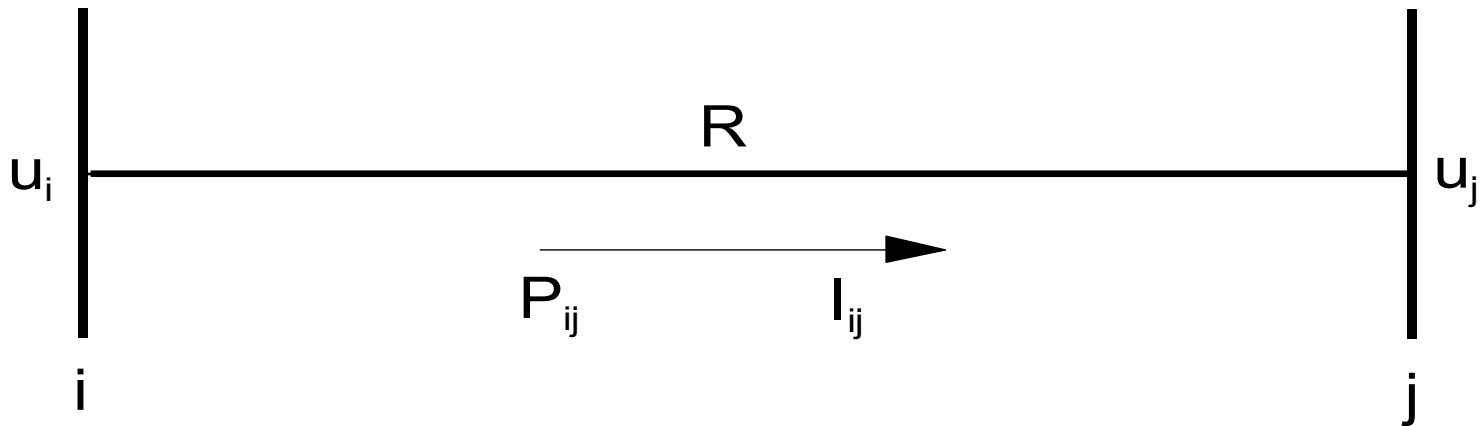
Take second equation:

$$\left. \begin{aligned} \frac{\lambda_1}{\lambda_2} &= \frac{(2U_2 - U_1)/R}{U_1/R} = 2\frac{U_2}{U_1} - 1 \\ \frac{U_2}{U_1} &= 1 - \frac{\rho_{12}}{2} \end{aligned} \right\} \Rightarrow \frac{\lambda_1}{\lambda_2} = 2\left(1 - \frac{\rho_{12}}{2}\right) - 1 = 1 - \rho_{12} \Rightarrow \rho_{12} = 1 - \frac{\lambda_1}{\lambda_2} = \frac{\lambda_2 - \lambda_1}{\lambda_2}$$

and in general

$$\frac{\lambda_i}{\lambda_j} = 1 - \rho_{ij}$$

Transmission Losses



- In unconstrained power system, differences between prices caused by grid losses alone
- ρ_{ij} is the marginal transmission loss for transfer power from node i to node j

$$\frac{\lambda_i}{\lambda_j} = 1 - \rho_{ij}$$

3-node DC grid: Flows and Loss coefficients (uncongested)

Flows:

$$P_{12} = U_1 I_{12} = U_1 \frac{U_1 - U_2}{R_{12}} = 61.89 \times \frac{61.89 - 62.44}{10} = -3.40 \text{ MW}$$

Marginal Transmission Loss:

$$\rho_{12} = 1 - \frac{\lambda_1}{\lambda_2} = 1 - \frac{14.247}{14} = -1.8\% \quad \rho_{21} = +1.7\%$$

$$\rho_{13} = 1 - \frac{\lambda_1}{\lambda_3} = 1 - \frac{14.247}{15.148} = +5.9\% \quad \rho_{31} = -6.3\%$$

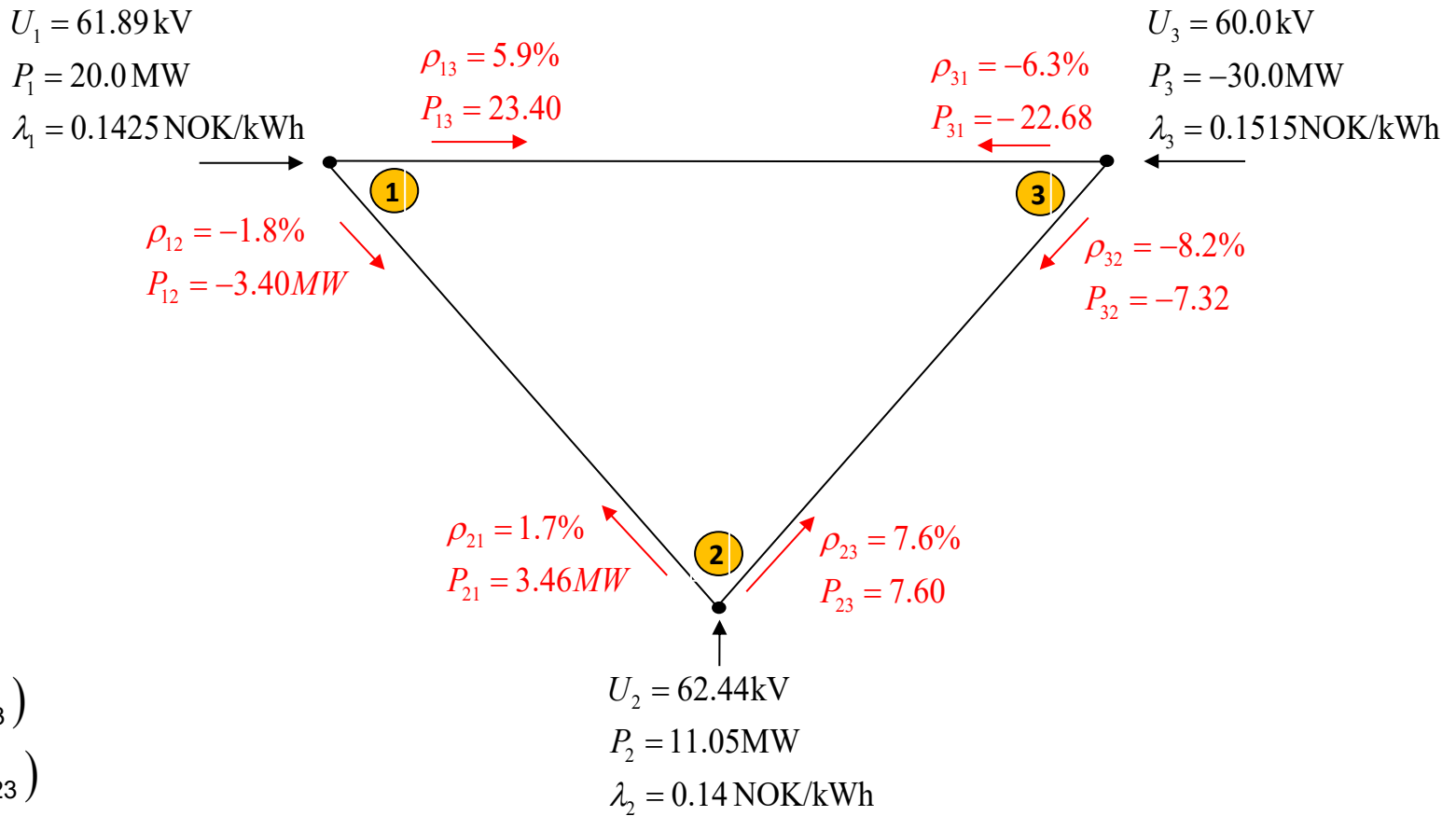
$$\rho_{23} = 1 - \frac{\lambda_2}{\lambda_3} = 1 - \frac{14}{15.148} = 7.6\% \quad \rho_{32} = -8.2\%$$

$$\lambda_1 = \lambda_3 (1 - \rho_{13}) = 15.15 (1 - 5.9\%) = 14.25 \quad [\text{øre/kWh}]$$

$$\lambda_2 = \lambda_3 (1 - \rho_{23}) = 15.15 (1 - 7.6\%) = 14 \quad [\text{øre/kWh}]$$

3-node DC grid

uncongested, marginal transmission losses and flows



$$\lambda_1 = \lambda_3 (1 - \rho_{13})$$

$$\lambda_2 = \lambda_3 (1 - \rho_{23})$$

$$P_{loss} = \sum_i P_{Gi} - Load = 31.05 - 30 = 1.05 \text{ MW}$$

3-node DC grid, Congested case

Set limit $I_{13} = 350$ A (uploaded document uses 21 MW at bus 3).

$$I_{13} = \frac{1}{R_{13}}(U_1 - U_3) \leq I_{13\max} \Rightarrow (U_1 - U_3) \leq 5 \times 350 = 1.75 \text{ kV} \Rightarrow (U_1 - U_3) - 1.75 \leq 0$$

$$P_{31} = \frac{U_3(U_3 - U_1)}{R_{13}} \leq 21 \text{ MW} \Rightarrow \frac{U_3(U_3 - U_1)}{R_{13}} - 21 \leq 0$$

$$\mathbf{h}(\mathbf{U}) \leq 0 \Rightarrow \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = \begin{bmatrix} |I_{13}| - I_{13,\max} \\ |I_{12}| - I_{12,\max} \\ |I_{23}| - I_{23,\max} \end{bmatrix} \leq 0 \Rightarrow \begin{bmatrix} |0.2(U_1 - U_3)| - I_{13,\max} \\ |0.1(U_1 - U_2)| - I_{12,\max} \\ |0.05(U_2 - U_3)| - I_{23,\max} \end{bmatrix} \leq 0$$

$$\frac{\partial \mathbf{h}}{\partial \mathbf{U}} = \begin{bmatrix} \frac{\partial h_1}{\partial U_1} & \frac{\partial h_1}{\partial U_2} \\ \frac{\partial h_2}{\partial U_1} & \frac{\partial h_2}{\partial U_2} \\ \frac{\partial h_3}{\partial U_1} & \frac{\partial h_3}{\partial U_2} \end{bmatrix} = \begin{bmatrix} 0.2 & 0 \\ 0.1 & -0.1 \\ 0 & 0.05 \end{bmatrix} \Rightarrow \frac{\partial \mathbf{h}}{\partial \mathbf{U}}^t \boldsymbol{\eta} = \begin{bmatrix} 0.2 & 0.1 & 0 \\ 0 & -0.1 & 0.05 \end{bmatrix} \begin{bmatrix} \eta_{13} \\ \eta_{12} = 0 \\ \eta_{23} = 0 \end{bmatrix} = \begin{bmatrix} 0.2\eta_{13} \\ 0 \end{bmatrix}$$

3-node DC grid, Congested case

General Equations:

$$g(\mathbf{U}, \mathbf{P}) = 0 \Rightarrow \begin{bmatrix} 0.3U_1^2 - 0.1U_2U_1 - 0.2 \cdot 60 \cdot U_1 - P_1 \\ -0.1U_1U_2 + 0.15U_2^2 - 0.05U_2 \cdot 60 - P_2 \\ -0.2U_1 \cdot 60 - 0.05U_2 \cdot 60 + 0.25 \cdot 60^2 - (-30) \end{bmatrix} = 0 \quad (a)$$

$$\mathbf{h}(\mathbf{U}) \Rightarrow (U_1 - U_3) = 1.75 \text{ kV} \quad (b)$$

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}} \right)^t \boldsymbol{\lambda} + \left(\frac{\partial \mathbf{h}}{\partial \mathbf{U}} \right)^t \boldsymbol{\eta} = 0 \Rightarrow \begin{bmatrix} \lambda_1(0.6U_1 - 0.1U_2 - 0.2 \cdot 60) - 0.1U_2\lambda_2 - 0.2 \cdot 60\lambda_3 + 0.2\eta_{13} \\ -0.1U_1\lambda_1 + \lambda_2(-0.1U_1 + 0.3U_2 - 0.05 \cdot 60) - 0.05 \cdot 60\lambda_3 \end{bmatrix} = 0 \quad (c)$$

$$\begin{aligned} \lambda_1 &= 12 \\ \lambda_2 &= 14 \end{aligned} \quad (d)$$

γ_1 and γ_2 are zero because $P_{1,\max}$ and $P_{2,\max}$ are not binding.

From (b) solve $U_1 = 61.75 \text{ kV}$ (U_3 given)

Then (a) become 3 linear equations with unknowns U_2 , P_1 , P_2 . Solve U_2 from last, then P_1 from first and P_2 from second

3-node DC grid, Congested case

$$U_2 = 63 \text{ kV}$$

$$P_1 = 13,894 \text{ MW}$$

$$P_2 = 17,325 \text{ MW}$$

The last equation in (c) solves λ_3 and finally first equation in (c) solves η_{13}

$$\lambda_3 = 20,683 \text{ øre/kWh}$$

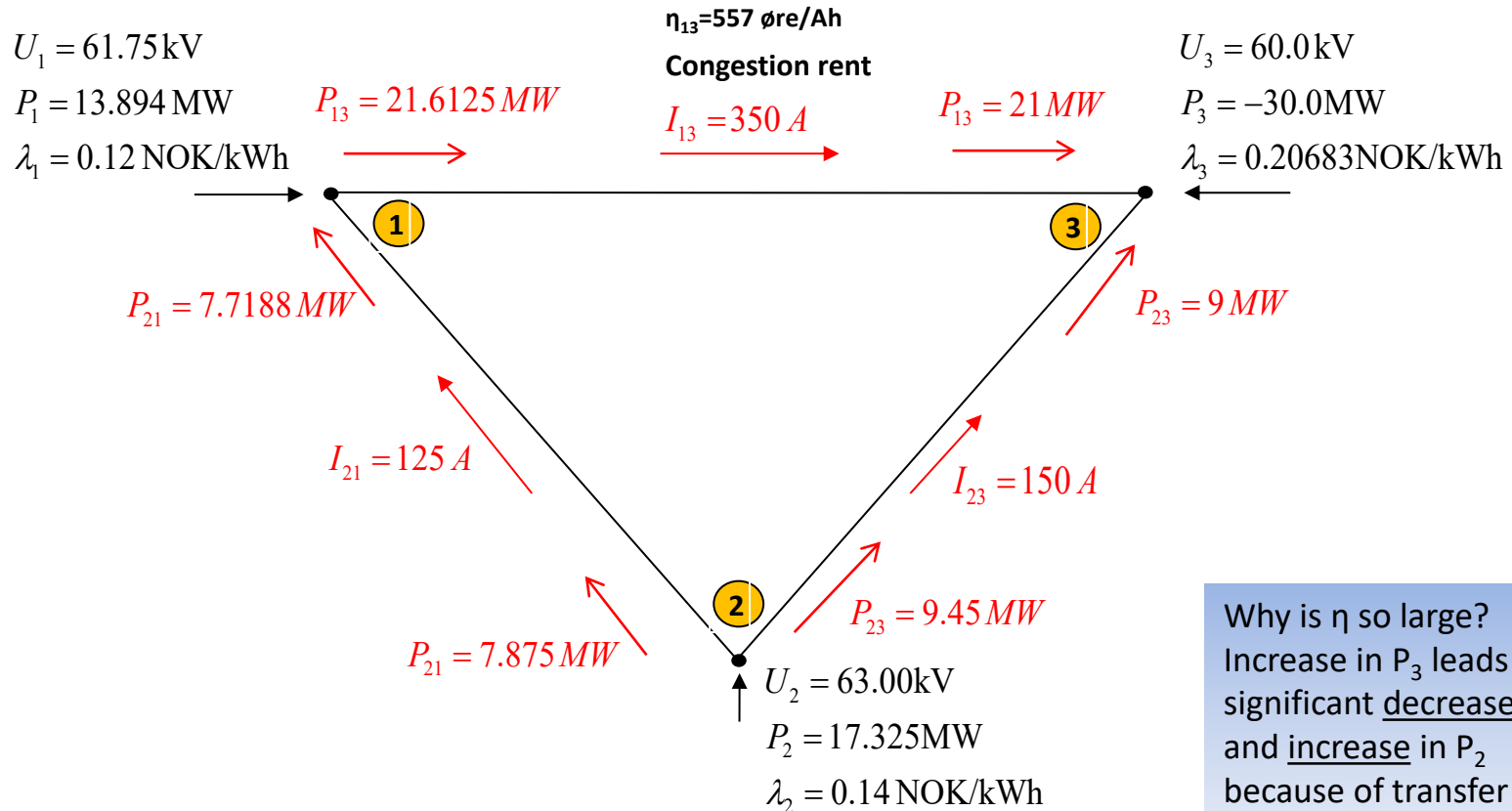
$$\eta_{13} = 557 \text{ øre/Ah}$$

If the constraint refers to the transmitted power to bus 3

$$\eta_{13,kWh} = \frac{557}{60} \left(\frac{\text{øre}}{\text{Ah} \times \text{kV}} \right) = 9.283 \text{ øre/kWh}$$

$$P_{loss} = \sum_i P_{Gi} - Load = 1,219 \text{ MW} \quad \text{Increased compare to uncongested case}$$

3-node DC grid, Congested case



$$P_{13} = I_{3\max} \cdot U_3 = 350 \times 60000 = 21 \text{ MW}$$

$$\eta_{13} = \frac{\partial f}{\partial P_{13\max}} = \frac{557}{60} \text{ øre}/(\text{Ah} \times \text{kV}) = 9.283 \text{ øre} / \text{kWh}$$

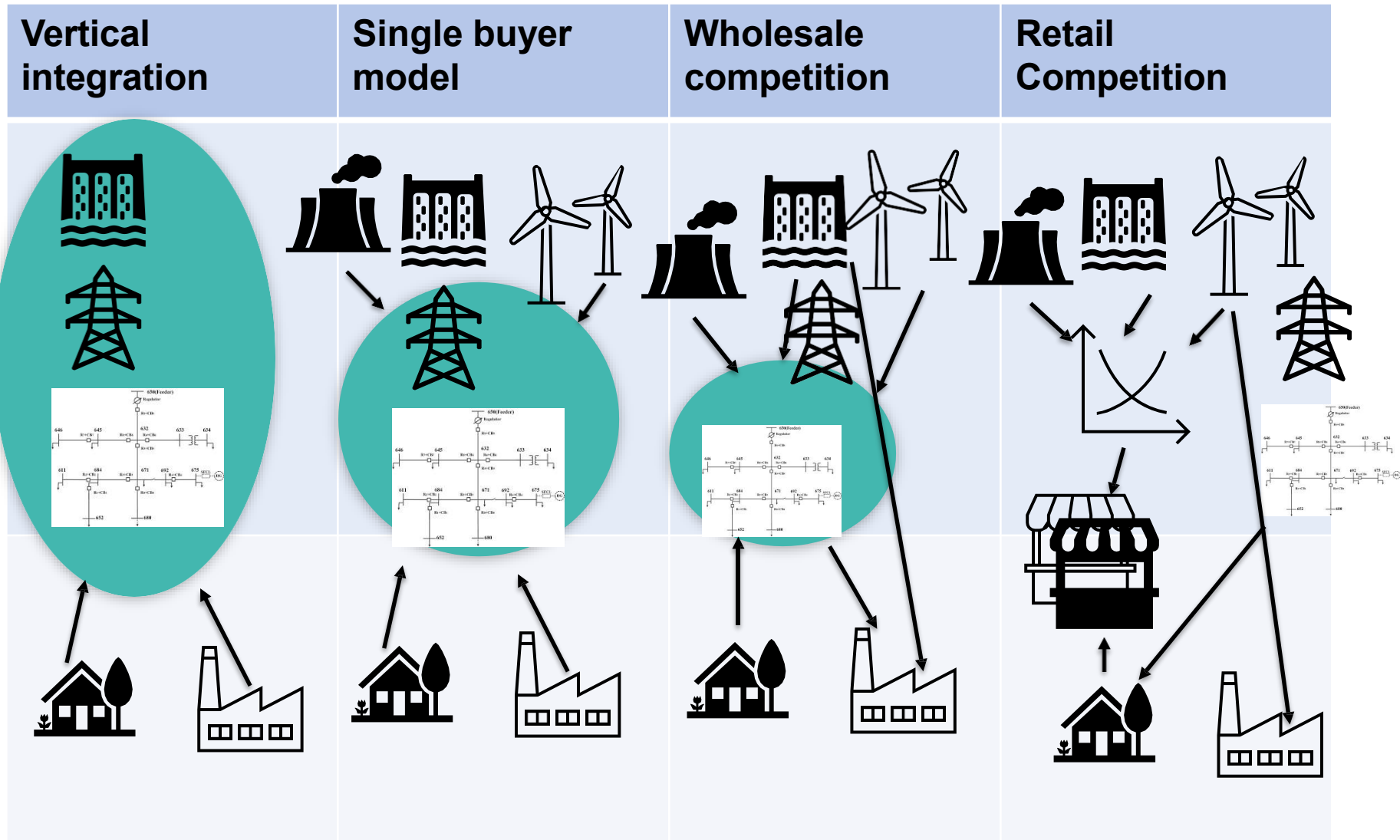
$$\text{Losses} = 1.219 \text{ MW}$$

How Marginal Loss Factors Are Calculated in Norway

- Norway does not implement nodal pricing, but:
 - It measures how much total system losses increase if injection at node i increases by 1 MW.
 - It is calculated at each transmission connection point.
 - All participants within the same bidding zone (NO1–NO5) face the same day-ahead spot price, regardless of their exact network location
- Norway internalizes marginal transmission losses through tariff settlement rather than nodal spot price differentiation.
 - Settlement: $\text{Loss Charge}_i = \rho_i \cdot P_i \cdot \lambda_{\text{zone}}$
- Marginal loss factor in Norway = derivative of total system losses with respect to injection at a connection point and representative operating scenarios, capped at $\pm 15\%$.

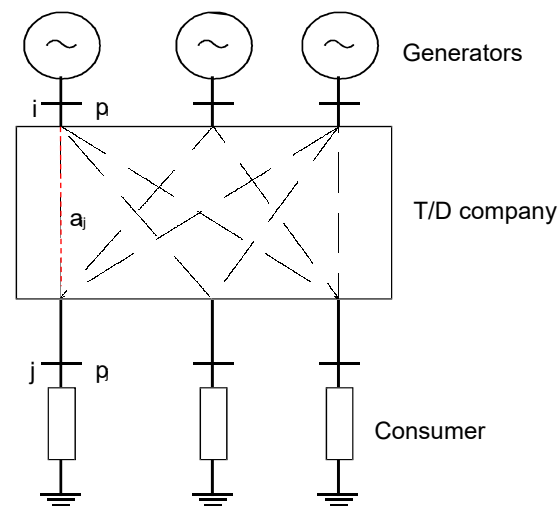
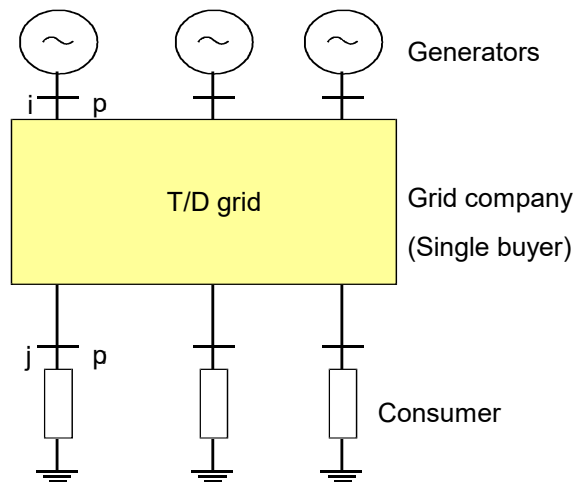
<https://www.statnett.no/en/for-stakeholders-in-the-power-industry/tariffs/how-we-calculate-the-tariff/>
<https://www.nve.no/norwegian-energy-regulatory-authority/network-regulation/network-tariffs/>

Different stages of grid openness

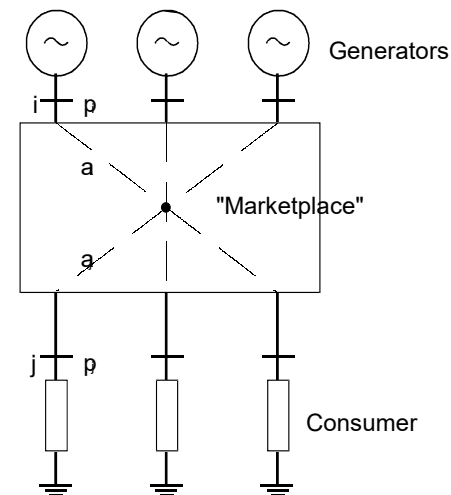


Nodal prices and transmission pricing

- Single Buyer
- Point-to-point transmission pricing
- Point tariffs system



$$a_{ij} = p_j - p_i = (\lambda_j - \lambda_i)$$



$$a_i = p_s - \lambda_i$$

$$a_j = \lambda_j - p_s$$

Point tariffs and hub location

- Point tariff:
 - Producers pay for transmission from input node to "market hub"
 $a_i = p_s - \lambda_i$
 - they get paid $p_s - a_i$
 - Consumers pay for transmission from "market hub" to output node
 $a_j = \lambda_j - p_s$
 - they pay $p_s + a_j$
- Does the location of the hub influence the market?
 - yes
 - increase in grid tariff Δp on input nodes raises marginal cost of generators, increases market price with Δp ; $p_s + \Delta p$
 - but if grid tariff is decreased correspondingly with Δp at output nodes ($p_s - \Delta p$), sum of market price and grid tariff is unchanged for consumers

Two cases with respect to hub location

	Reference case	Δp case
Input from generators	P_i	P_i
Output of consumer	P_j	P_j
Transmission price input	a_i	$a_i + \Delta p$
Transmission price output	a_j	$a_j - \Delta p$
Market price	p_s	$p_s + \Delta p$
Grid losses	T	T
Generation cost	C_i	C_i

The losses are the difference between input and output:

$$T = \sum_{i=1}^n P_i - \sum_{j=1}^m P_j$$

Generating company	Reference case	Δp case
Generator sales income	$P_i p_s$	$P_i (p_s + \Delta p)$
Generator transmission cost	$P_i a_i$	$P_i (a_i + \Delta p)$
Generation cost	C_i	C_i
Generation surplus	$P_i (p_s - a_i) - C_i$	$P_i (p_s - a_i) - C_i$

Consumer	Reference case	Δp case
Market purchase	$P_j p_s$	$P_j (p_s + \Delta p)$
Transmission cost	$P_j a_j$	$P_j (a_j - \Delta p)$
Total expenses	$P_j (p_s + a_j)$	$P_j (p_s + a_j)$

Transmission company	Reference case	Δp case
Purchase of energy (losses)	$T p_s$	$T (p_s + \Delta p)$
Revenue: From generator	$\sum P_i a_i$	$P_i (a_i + \Delta p)$
From consumers	$\sum P_j a_j$	$P_j (a_j - \Delta p)$
Increased cost: $T \Delta p$		
Increased revenue: $\sum P_i \Delta p - \sum P_j \Delta p = \Delta p (\sum P_i - \sum P_j) = T \Delta p$		

Conclusion hub location

- Market price changes
- Grid tariff changes
- but no net difference for market participants!
 - producers
 - consumers
 - grid company

Market Settlement: Uncongested case:

Placing hub in **node 3**

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}}\right)' \boldsymbol{\lambda} = \begin{bmatrix} 18.89\lambda_1 - 6.244\lambda_2 - 12.00\lambda_3 \\ -6.189\lambda_1 + 9.543\lambda_2 - 3.00\lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda'_3 = 1 \Rightarrow \begin{bmatrix} 18.89\lambda'_1 - 6.244\lambda'_2 - 12.00 \\ -6.189\lambda'_1 + 9.543\lambda'_2 - 3.00 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $\lambda'_1 = 0.941$ and $\lambda'_2 = 0.924$

The corresponding marginal loss factors become

$$\rho_1 = \rho_{1s} = \frac{\lambda_s - \lambda_1}{\lambda_s} = 1 - \lambda'_1 = 0.059$$

$$\rho_2 = \rho_{2s} = \frac{\lambda_s - \lambda_2}{\lambda_s} = 1 - \lambda'_2 = 0.076$$

$$\rho_3 = \rho_{3s} = 0$$

The system price at node 3: $\lambda_3 = 15.148$ øre/kWh

$$\lambda_1 = \lambda_s - \rho_1 \lambda_s = 15.148 - 0.059 \times 15.148 = 14.25 \text{ øre/kWh}$$

$$\lambda_2 = \lambda_s - \rho_2 \lambda_s = 15.148 - 0.076 \times 15.148 = 14 \text{ øre/kWh}$$

Market Settlement: Uncongested case:

Placing hub in **node 3**

The economic consequences for the different entities will be :

Generator 1:		
Sales:	$P_1 \cdot \lambda_s = 20\text{MW} \times 15.148\text{øre/kWh}$	=3030 NOK/h
-Transmission Cost:	$P_1 \cdot \rho_1 \cdot \lambda_s$	=179 NOK/h
Net revenue:		=2851 NOK/h

Generator 2:		
Sales:	$P_2 \cdot \lambda_s = 11.05\text{MW} \times 15.148\text{øre/kWh}$	=1674 NOK/h
-Transmission Cost:	$P_2 \cdot \rho_2 \cdot \lambda_s$	=127 NOK/h
Net revenue:		=1547 NOK/h

Load 3:		
Purchase cost:	$P_3 \cdot \lambda_s = 30\text{MW} \times 15.148\text{øre/kWh}$	=4544 NOK/h
+Transmission Cost:	$P_3 \cdot \rho_3 \cdot \lambda_s$	=0 NOK/h
Net revenue:		=-4544 NOK/h

Market Settlement: Uncongested case:

Placing hub in **node 3**

Transmission company		
Income from Generator 1		=179 NOK/h
Income from Generator 2		=127 NOK/h
Income from load		=0 NOK/h
-Transmission losses	$P_{\text{loss}} \cdot \lambda_s = 1.05 \text{ MW} \times 15.148 \text{ øre/kWh}$	=159 NOK/h
Net revenue		=147 NOK/h

Market Settlement: Uncongested case:

Placing hub in **node 2**

$$\left(\frac{\partial \mathbf{g}}{\partial \mathbf{U}}\right)' \boldsymbol{\lambda} = \begin{bmatrix} 18.89\lambda_1 - 6.244\lambda_2 - 12.00\lambda_3 \\ -6.189\lambda_1 + 9.543\lambda_2 - 3.00\lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\lambda'_2 = 1 \Rightarrow \begin{bmatrix} 18.89\lambda'_1 - 6.244 - 12.00\lambda'_3 \\ -6.189\lambda'_1 + 9.543 - 3.00\lambda'_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives $\lambda'_1 = 1.018$ and $\lambda'_3 = 1.082$

The corresponding marginal loss factors become

$$\rho_1 = \rho_{1s} = \frac{\lambda_s - \lambda_1}{\lambda_s} = 1 - \lambda'_1 = -0.018$$

$$\rho_2 = \rho_{2s} = 1 - 1 = 0$$

$$\rho_3 = \rho_{s3} = \frac{\lambda_3 - \lambda_s}{\lambda_s} = \lambda'_3 - 1 = 0.082 \quad \text{Exchange from hub to node 3}$$

The system price at node 2: $\lambda_2 = 14$ øre/kWh

$$\lambda_1 = \lambda_s - \rho_1 \lambda_s = 14 - (-0.018) \times 14 = 14.25 \text{ øre/kWh}$$

$$\lambda_3 = \lambda_s + \rho_3 \lambda_s = 14 + 0.082 \times 14 = 15.148 \text{ øre/kWh}$$

Market Settlement: Uncongested case:

Placing hub in **node 2**

The economic consequences for the different entities will be:

Generator 1:		
Sales:	$P_1 \cdot \lambda_s = 20\text{MW} \times 14\text{øre/kWh}$	=2800 NOK/h
-Transmission Cost:	$P_1 \cdot \rho_1 \cdot \lambda_s = 20 \times (-0.018) \times 14$	=-50 NOK/h
Net revenue:		=2850 NOK/h

Generator 2:		
Sales:	$P_2 \cdot \lambda_s = 11.05\text{MW} \times 14\text{øre/kWh}$	=1547 NOK/h
-Transmission Cost:	$P_2 \cdot \rho_2 \cdot \lambda_s$	=0 NOK/h
Net revenue:		=1547 NOK/h

Load 3:		
Purchase cost:	$P_3 \cdot \lambda_s = 30\text{MW} \times 14\text{øre/kWh}$	=4200 NOK/h
+Transmission Cost:	$P_3 \cdot \rho_3 \cdot \lambda_s$	=344 NOK/h
Net revenue:		=-4544 NOK/h

Market Settlement: Uncongested case:

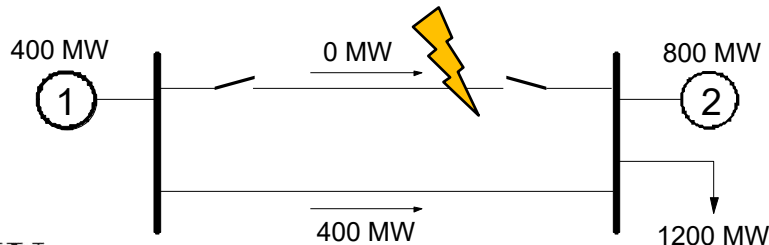
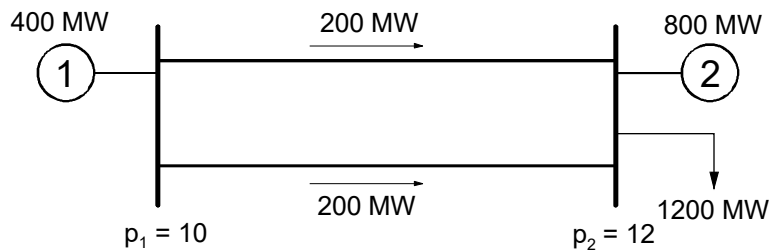
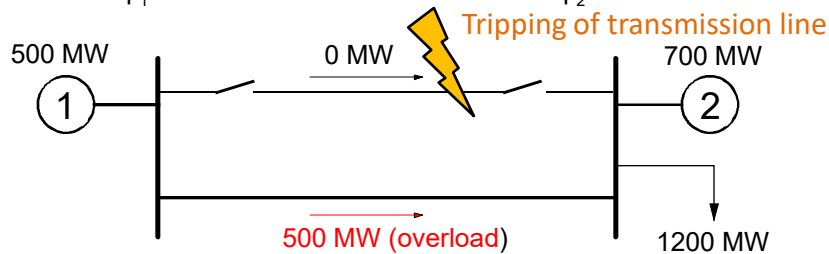
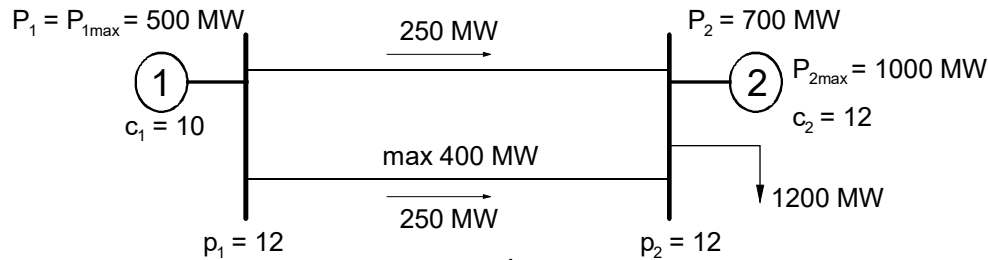
Placing hub in **node 2**

Transmission company		
Income from Generator 1		=-50 NOK/h
Income from Generator 2		=0 NOK/h
Income from load		=344 NOK/h
-Transmission losses	$P_{\text{loss}} \cdot \lambda_s = 1.05 \text{ MW} \times 14 \text{ øre/kWh}$	=147 NOK/h
Net revenue		=147 NOK/h

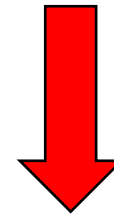
3-node DC grid example, Cash flow uncongested case

	Hub at node 3	Hub at node 2
Spot price p_s (NOK/kWh)	0.15148	0.140
Nodal "loss" factors:		
ρ_{1in}	0.059	-0.018
ρ_{2in}	0.076	0.0
ρ_{3out}	0.0	0.082
Generator 1		
Sales $P_1 p_s$ (NOK/h)	3030	2800
- transmission $P_1 \rho_{1in} p_s$ (NOK/h)	- 179	+ 50
Net income (NOK/h)	2851	2850
Generator 2		
Sales $P_2 p_s$ (NOK/h)	1674	1547
- transmission $P_2 \rho_{2in} p_s$ (NOK/h)	- 127	0
Net income	1547	1547
Load		
Purchase $P_3 p_s$ (NOK/h)	4544	4200
- transmission $P_3 \rho_{3out} p_s$ (NOK/h)	0	344
Total	4544	4544
Grid company		
Transmission gen 1 (NOK/h)	179	-50
Transmission gen 2 (NOK/h)	127	0
Transmission load (NOK/h)	0	344
- losses $P_{loss} \cdot p_s$ (NOK/h)	- 159	- 147
Net revenue (NOK/h)	147	147

Security constrained OPF (SCOPF)



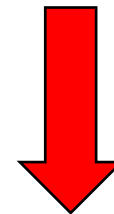
Optimal dispatch



Post-contingency state (brownout)



Secure dispatch



Secure post-contingency state

Recap

- Lagrange multipliers – short repetition
- OPF equations based on Lagrange multiplier approach
- Example using 3-node DC network with resistances
- Nodal prices, losses, market hub location, security of supply